# AFRID PASHA

AI & Data Science Engineer

+91-9008938007 | pashaafrid175@gmail.com

github.com/Afrid-glitch | linkedin.com/in/afrid-pasha-bbb726315

## ABOUT ME

I'm Afrid Pasha, a third-year B.Tech student in Artificial Intelligence and Data Science at Visvesvaraya Technological University, Mysore, with a CGPA of 8.48/10. I'm passionate about turning data into intelligent, real-world solutions.

My technical foundation covers Python, SQL, and core ML algorithms, along with Deep Learning (ANN, CNN, RNN, Transformers) and NLP. I also work with Generative AI and Agentic AI, building RAG pipelines with ChromaDB, LangChain-based agents, and MCP integrations. On the data side, I'm comfortable with Pandas, NumPy, EDA, and Power BI for insight-driven dashboards. To take models from notebook to production, I use FastAPI, Streamlit, Docker, Kubernetes, and Git, following MLOps best practices.

I'm a quick learner who enjoys solving practical problems and keeping up with the fast-moving AI landscape. I'm currently seeking internship and entry-level opportunities where I can apply my skills, contribute to impactful work, and grow alongside a strong engineering team.

## TECHNICAL SKILLS

### 1. Programming & Web Development

- **Python:** Core language for data science, AI/ML and automation; OOP, file handling, APIs and scripting.
- **C:** Strong programming fundamentals including data structures, pointers and memory handling.
- **JavaScript:** Dynamic, interactive web functionality and asynchronous programming.
- **HTML & CSS:** Semantic markup and responsive, well-styled web page layouts.

### 2. Databases

- **PostgreSQL:** Relational database design, writing SQL queries, joins, aggregations and data retrieval.
- **MongoDB:** NoSQL document database for flexible, schema-less data storage in full-stack applications.

### 3. Machine Learning

- **Linear Regression:** Predicting continuous values by modelling relationships between variables.
- **Logistic Regression:** Binary and multi-class classification using probability-based decision boundaries.
- **K-Nearest Neighbours (KNN):** Distance-based classification and regression.
- **Decision Tree:** Rule-based, interpretable models for classification and regression.
- **Random Forest:** Ensemble learning that combines many trees for higher accuracy and reduced overfitting.

### 4. Deep Learning

- **ANN (Artificial Neural Networks):** Fully connected networks for prediction and pattern learning.
- **CNN (Convolutional Neural Networks):** Image classification and computer vision tasks.
- **RNN (Recurrent Neural Networks):** Sequential and time-series data modelling.
- **Transformers:** Attention-based architectures powering modern NLP and Generative AI.

### 5. Natural Language Processing (NLP)

- Text preprocessing, tokenization, stemming and lemmatization, and text classification.
- Sentence embeddings and semantic similarity, sentiment analysis and text summarization.
- Working with pre-trained HuggingFace Transformer models.

### 6. Generative AI & Agentic AI

- **Generative AI & LLMs:** Large Language Models, prompt engineering, text generation and summarization.
- **RAG (Retrieval-Augmented Generation):** Grounding LLM answers in custom data using embeddings and vector databases (ChromaDB).
- **Agentic AI:** Building autonomous agents that plan, reason, use tools and collaborate in multi-agent workflows.
- **MCP (Model Context Protocol):** Connecting AI models to external tools, data sources and services in a standard way.
- **LangChain & LCEL:** Chains, ReAct agents, agent executors and tool integration.
- **Related concepts:** Embeddings, vector search, function/tool calling, memory, fine-tuning basics, AI guardrails and evaluation.

### 7. Data Science

- **NumPy:** Fast numerical computing and array operations.
- **Pandas:** Data manipulation, cleaning, merging and analysis of structured data.
- **Matplotlib & Seaborn:** Statistical and exploratory data visualization.
- **EDA Process:** Data cleaning, handling missing values and outliers, feature analysis, correlation study and uncovering insights.

### 8. Data Analytics

- **MS Excel:** Formulas, pivot tables, data cleaning and reporting.
- **Power BI:** Interactive dashboards, data modelling and business intelligence reports.

### 9. Tools, Deployment & MLOps

- **Git & GitHub:** Version control, branching and collaborative development.
- **FastAPI:** Building fast REST APIs to serve applications and ML models.
- **Streamlit & Streamlit Cloud:** Building and hosting interactive data and AI web apps.
- **Render:** Cloud deployment of web applications and APIs.
- **Docker:** Containerizing applications for consistent, portable deployment.
- **Kubernetes:** Container orchestration, scaling and management.
- **MLOps Concepts:** ML lifecycle management, model versioning, CI/CD pipelines, monitoring and reproducible deployment.
- **VS Code:** Primary development environment.

## EDUCATION

**B.Tech, Artificial Intelligence and Data Science** (3rd Year) **CGPA: 8.48 / 10**
Visveswaraya Technological University, Mysore

**12th Grade (PCMB)** **91%**
Lions Pre University, Kollegal

**SSLC (10th Grade)** **91%**
Bird Memorial High School, Kollegal